Layouting of the comment tree

1 Introduction

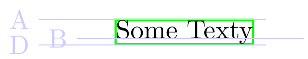
This document describes how the layouting of he comment tree is done. A comment tree is a (possibly nested) tree of either text, bracket, note or grid nodes.

In the following pictures, bounding boxes are drawn in green. Alignment lines are drawn in blue. Helper annotations are drawn in red. Hatched rectangles are objects that are handled as a black box, like sub grids.

2 Leafs

There are the terminal nodes of the comment tree. They don't have any subnodes.

2.1 Text



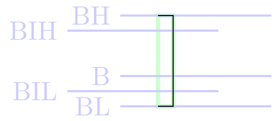
Text nodes have 3 main lines used for aligning them. The base line `B` is just the baseline of the text.

The line `A` is the ascender of the text.

The line `D` is the descender of the text.

Text nodes bounding box is always as heigh and wide as the text.

2.2 Bracket

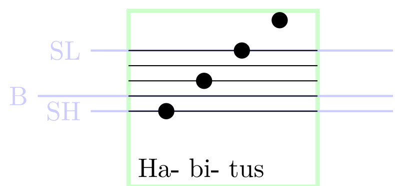


`BL` and `BH` denote the lowest and highest point of the bracket, respectively.

Brackets always have the same width. Brackets don't have a fixed height but get their height based on their neighboring nodes in the same grid row. The neighbors request a height and the height requested ist always the height between `BIL` and `BIH`. There is a default margin between `BIH` and `BH`, as well as between `BIL` and `BL`.

The baseline `B` is variable and depends on the neigbhours.

2.3 Notes



The bounding box of the notes is the bounding box of all the text on the bottom as well as the top notes.

There are three main lines used for alignment. The baseline `B` is always the second staff line from the bottom. The other two import lines are `SL` and `SH` which denote the lowest and highest staff line, respectively.

3 Grid

Grid nodes allow the layout of sub nodes. They can also themselves be the children of other grid nodes. In this case, they are handled as a black box with no special alignment lines.

3.1 Horizontal

The horizontal alignment of the grids is fairly simple. Their bouding box is used and centered on the column by default. If the node has a justification (i.e. left, center, right), the bounding box is shifted accordingly.

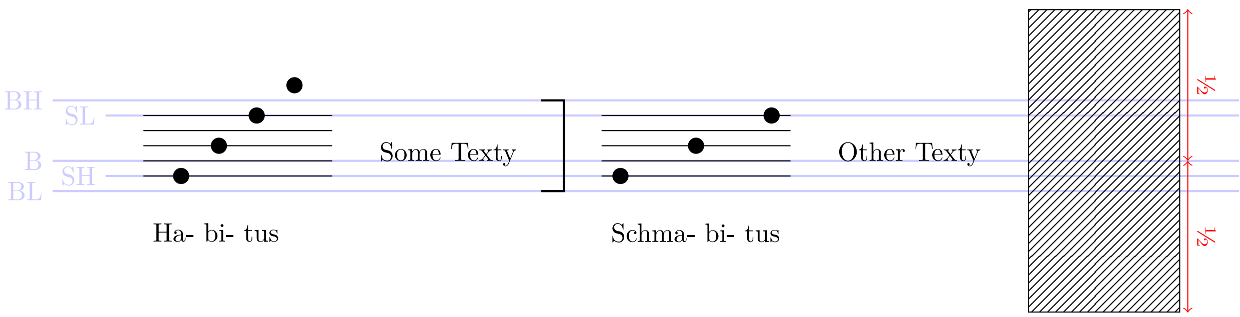
3.2 Vertical

The vertical alignment of nodes inside a row is harder. There are basically three cases which depend on what the dominating node in the row is.

3.2.1 Note dominated

If there are one or more note-nodes in the row, one of them will be the dominating node. It doesn't matter which one since all of them have the same alignment lines after they are themselves aligned via their baselines.

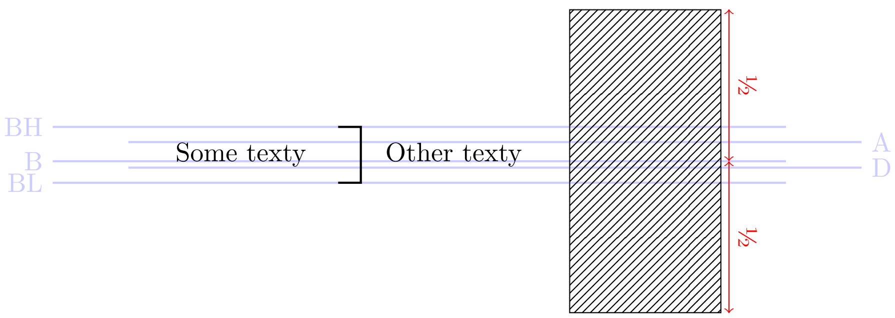
All notes will be aligned according to the baseline of the dominating node. In this case, other notes will be aligned by their second staff lines. Text nodes will be aligned by their baseline. Opaque nodes like sub grids will be aligned by the vertical center of their respective bounding boxes. Brackets will be generated with the inner height (`BIL` to `BIH`) set to `SL` to `SH`. The baseline will be set to one fourth the way between `BIL` and `BIH`.



3.2.2 Text dominated

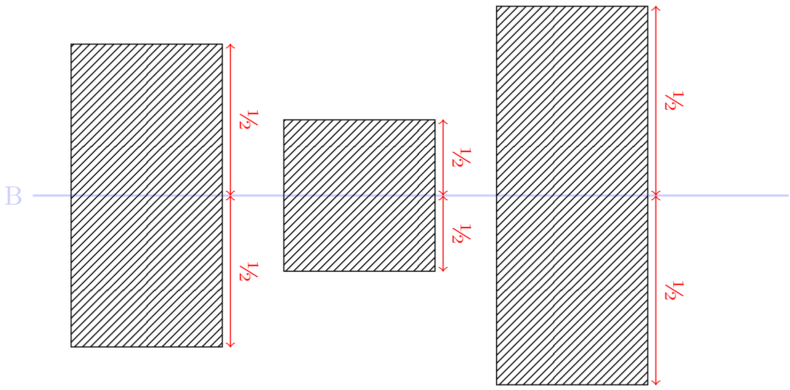
In case there are no note-nodes in the row, text nodes will be the dominating. Which one, again, doesn't matter, since they are aligned via baseline and then have the same ascender/descener lines after that.

The layouting will work as above, only brackets are generated differently. The inner height of the bracket (`BIL` to `BIH`) will be set to the height of the text nodes in the row, i.e. to `D` to `A`. The baseline will be $\frac{B-A}{D-A}$ the way between `BIL` and `BIH`.



3.2.3 Undominated

If only opqaue nodes are left in the row, they will all be aligned by their vertical center. Brackets will always have some default height, left as an implementation detail.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